Preface

Spring AI makes it possible for Java developers to build intelligent applications using the familiar Spring programming model.

This book helps Java and Spring developers move from AI curiosity to working applications. Through practical projects, you'll learn how to use Spring AI to build, test, observe, and extend intelligent systems with confidence.

In this book, you will learn how to:

- Understand core AI concepts and Spring AI abstractions
- Build Java applications with chat and multimodal models
- Switch AI providers and local models with minimal changes
- Structure prompts, outputs, and external data effectively
- Add speech, transcription, and image capabilities to applications
- Evaluate responses, monitor usage, and manage AI costs
- Extend Spring AI for advanced and custom use cases

Written by John Blum, a staff software engineer formerly with Spring R&D at VMware and a contributor to Spring AI, Spring Boot, Spring Data, Spring Framework, and Spring Session, this book combines deep Spring expertise with hands-on AI experience.

Learn how to turn AI ideas into practical Java applications without fighting unfamiliar tools or excessive boilerplate. This book shows you how to use Spring AI to build intelligent applications with the Spring programming model, helping you work productively while applying chat, audio, image, embeddings, structured output, and automation capabilities in real projects.

You start with the foundations of AI and Spring AI, then move into building applications that translate messages in real time, process audio, identify songs, and compare model reasoning in game-based scenarios.

You also explore model options, observability, testing, extensions, and autonomous workflows so you can move beyond prototypes and build reliable applications. By the

end of this book, you will be comfortable using Spring AI to design, develop, evaluate, and refine intelligent Java applications for a wide range of use cases.

Who this book is for

This book is for Java developers, Spring developers, software engineers, and application architects who want to build practical AI-powered applications with Spring AI. Readers should be comfortable building applications with Java and Spring. Prior experience with AI is helpful but not required.

What this book covers

Chapter 1, *Getting Started with Spring AI*, introduces the foundations of building intelligent Java applications with Spring AI. You will learn how to create your first Spring AI application with Spring Boot, connect to different AI providers and models, and switch between cloud and local AI runtimes such as OpenAI, Google Gemini, Amazon Bedrock, and Ollama.

Chapter 2, *Understanding Spring AI: Part 1*, introduces the core concepts of AI and the foundational Spring AI APIs used to interact with different AI models. You will explore chat models, advisors, embeddings, moderation, image generation, and audio transcription while learning how Spring AI provides a consistent programming model across multiple AI providers.

Chapter 3, *Understanding Spring AI: Part 2*, continues exploring Spring AI by focusing on more advanced APIs and application patterns. You will learn about streaming interactions, prompt engineering, structured output, document processing, and function calling, enabling you to build more interactive, context-aware, and intelligent applications.

Chapter 4, *Real-Time Message Translation*, shows how to build a multilingual messaging application using Spring AI and modern AI models. You will learn how to implement real-time language translation, speech-to-text, and text-to-speech capabilities while integrating AI-powered features into a modern chat application.

Chapter 5, *Beat Shazam*, explores how to build a music recognition application inspired by Shazam using Spring AI. You will learn how to process audio with the Java Sound API, generate audio fingerprints, store embeddings in a vector database, and identify songs using similarity search. The chapter also demonstrates how Spring AI can be extended to

support advanced, real-world AI use cases involving audio processing and vector-based retrieval.

Chapter 6, *Connecting the Dots*, demonstrates how to use multiple AI models together in a Spring AI application through a competitive game of Connect 4. You will learn how to work with prompts, structured output, prompt templates, and multiple AI providers while exploring how different AI models reason, make decisions, and respond under game-like conditions. The chapter also highlights the importance of prompt engineering, data modeling, and structured interactions when building intelligent applications with Spring AI.

Chapter 7, *Become a Spring AI Power* User, explores advanced Spring AI capabilities for building more reliable, observable, and production-ready AI applications. You will learn how to tune AI models with chat and model options, track token usage and costs through metadata and observability, implement testing strategies for AI-generated content, and extend Spring AI with custom components, tools, and integrations to support real-world application development.

Chapter 8, *Secret Agents*, introduces the foundations of Agentic AI and demonstrates how to build AI agents using Spring AI and the Model Context Protocol (MCP). You will learn about common agentic workflows, tool calling, memory, and retrieval while building a practical travel assistant capable of searching for flights and hotels using natural language interactions. The chapter also explores MCP integration with Claude Desktop and provides an overview of other JVM-based agent frameworks, including Embabel, Koog, and LangChain4j.

Chapter 9, *Demystifying the Impact of AI*, reflects on the broader impact of AI on technology, software engineering, and society. The chapter explores the evolution of computing, the disruptive nature of AI-driven automation, the role of AI in software development, and the ethical challenges surrounding intelligent systems. It also discusses the future of AI, Agentic AI, and responsible innovation while encouraging developers to think critically about how AI shapes human experiences and the future of technology.

To get the most out of this book

This book uses an applied approach to learning about and building practical AI-based applications with JVM and Spring technologies. This book need not be read cover to cover. However, if you are new to AI, the opening chapters will give you a fundamental understanding of AI along with a solid foundation for using Spring AI effectively. Familiarity with the following technologies will help you make the most of your learning journey:

- Java, or similar JVM language, and the core libraries
- Basic data structures and algorithms
- Software Design Patterns, SOLID principles, SPIs, JSON and REST
- Core Spring concepts (e.g. Dependency Injection) Aspect-Oriented Programming (AOP), and Spring Boot convention over configuration
- RDBMS, such as PostgreSQL or MySQL, Vector Stores and Caching
- Docker environments
- Apache Maven
- ChatGPT, Claude Desktop, Ollama, and IDE (optional)

While familiarity and application of JVM and Spring technologies is essential, the rest of this list is helpful, but not required.

Finally, my hope is that you learn a few things along the way and have fun doing so. It is an exciting time to be a software engineer.

Download the example code files

The code bundle for the book is hosted on GitHub at https://github.com/PacktPublishing/Building-Intelligent-Applications-With-Spring-AI. We also have other code bundles from our rich catalog of books and videos available at https://github.com/PacktPublishing. Check them out!

Download the color images

We also provide a PDF file that has color images of the screenshots/diagrams used in this book. You can download it here: https://packt.link/gbp/9781836204619.

Conventions used

There are a number of text conventions used throughout this book.

`CodeInText` : Indicates code words in text, database table names, folder names, file-names, file extensions, pathnames, dummy URLs, user input, and Twitter handles. For example: " There is little reason why the `findById(..)` method would return a `nullSong` entity at this step in the code and throw a `SongNotFoundException` unless the "matching" `Song` was deleted prior to search."

A block of code is set as follows:

```
interface ChatModel extends Model<Prompt, ChatResponse>, StreamingChatModel {
  default String call(String message) { ... }
```

```
  default String call(Message... messages) { ... }
  ChatResponse call(Prompt prompt);
  ChatOptions getDefaultOptions();
  default Flux<ChatResponse> stream(Prompt prompt) { ... }
}
```

Any command-line input or output is written as follows:

```
user> Knock Knock
ai> Who's there?
user> Orange
ai> Orange who?
user> Orange you glad we met?
ai> That's a classic one! Well played! Would you like to play another round of
Knock Knock?
```

Bold: Indicates a new term, an important word, or words that you see on the screen. For instance, words in menus or dialog boxes appear in the text like this. For example: " Also, pay attention to other libraries and frameworks in the AI space, like **Embabel** and **Langchain4j**, as well as in other stacks, like Python."

NOTE

Warnings or important notes appear like this.

TIP

Tips and tricks appear like this.

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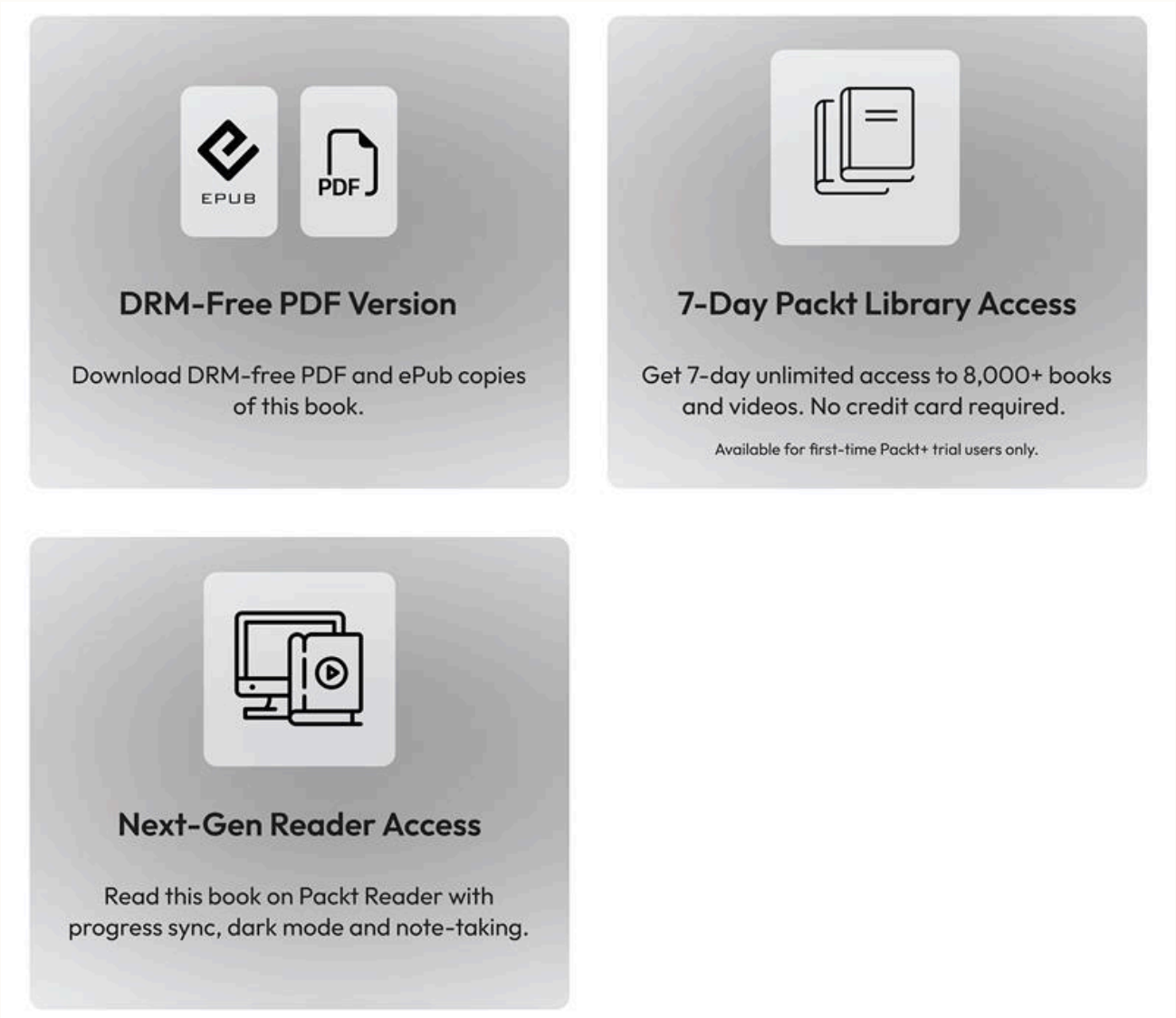
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